

HospitalDB

SQL Server Database Documentation

Mahmoud Atef | mahmoudeimo95@gmail.com | github.com/mahmoudeimo95

Database Overview

Tables	9
Total Sample Records	22+
Views	1 (vw_PatientHistory)
Stored Procedures	1 (AddPatient)
Database	SQL Server 2022

Entity Relationship Overview

Key: PK = Primary Key | FK = Foreign Key | UNIQUE = Unique Constraint | CHECK = Check Constraint

Table	Relates To	Relationship
Departments	Doctors	One department has many doctors
Departments	Appointments	One department has many appointments
Doctors	Appointments	One doctor has many appointments
Patients	Appointments	One patient has many appointments
Patients	MedicalRecords	One patient has many records
Patients	Bills	One patient has many bills
Appointments	Treatments	One appointment leads to treatments
Appointments	Bills	One appointment generates a bill
Treatments	Prescriptions	One treatment includes prescriptions

Table: Departments

Column	Type	Constraint	Description
DepartmentID	INT	PK IDENTITY	Auto-generated ID
DepartmentNM	VARCHAR(100)	NOT NULL UNIQUE	Department name
Location	VARCHAR(100)	-	Physical location
Phone	VARCHAR(20)	-	Contact number

Sample Data — Departments

ID	Name	Location	Phone
1	Cardiology	Building A - 2nd Floor	01000000001
2	Neurology	Building B - 1st Floor	01000000002
3	Orthopedics	Building C - 3rd Floor	01000000003

Table: Doctors

Column	Type	Constraint	Description
DoctorID	INT	PK IDENTITY	Auto-generated ID
DepartmentID	INT	FK -> Departments	Linked department
DoctorNM	VARCHAR(100)	NOT NULL	Full name
Specialization	VARCHAR(100)	-	Medical specialty
Phone	VARCHAR(20)	-	Contact number
Email	VARCHAR(100)	UNIQUE	Email address
HireDate	DATE	DEFAULT GETDATE()	Date hired

Sample Data — Doctors

ID	DeptID	Name	Specialization	Email
1	1	Dr. Ahmed Ali	Heart Specialist	ahmed@hospital.com
2	2	Dr. Sara Mohamed	Brain Specialist	sara@hospital.com
3	3	Dr. Omar Hassan	Bone Specialist	omar@hospital.com

Table: Patients

Column	Type	Constraint	Description
PatientID	INT	PK IDENTITY	Auto-generated ID
FullName	VARCHAR(100)	NOT NULL	Full name
Gender	VARCHAR(10)	CHECK ('Male','Female')	Gender
BirthDate	DATE	-	Date of birth
Phone	VARCHAR(20)	-	Contact number
BloodType	VARCHAR(5)	-	Blood type
Address	VARCHAR(200)	-	Home address
RegistrationDate	DATE	DEFAULT GETDATE()	Date registered

Sample Data — Patients

ID	Full Name	Gender	Birth Date	Blood	Address
1	Khaled Mostafa	Male	1995-05-10	O+	Cairo
2	Mona Adel	Female	2000-09-21	A+	Giza
3	Youssef Tarek	Male	1988-12-01	B-	Alexandria
4	Osman	Male	2004-06-15	NULL	NULL

Table: Rooms

Column	Type	Constraint	Description
RoomID	INT	PK IDENTITY	Auto-generated ID
RoomNumber	VARCHAR(10)	UNIQUE NOT NULL	Room number
RoomType	VARCHAR(50)	-	ICU / General / Surgery
FloorNumber	INT	-	Floor level
Status	VARCHAR(20)	CHECK ('Available','Occupied')	Room status

Sample Data — Rooms

ID	Number	Type	Floor	Status
1	101	ICU	1	Available
2	102	General	1	Occupied
3	201	Surgery	2	Available

Table: Appointments

Column	Type	Constraint	Description
AppointmentID	INT	PK IDENTITY	Auto-generated ID
PatientID	INT	FK -> Patients	Linked patient
DoctorID	INT	FK -> Doctors	Linked doctor
DepartmentID	INT	FK -> Departments	Linked department
AppointmentDate	DATETIME	NOT NULL	Date and time
Reason	VARCHAR(200)	-	Visit reason
Status	VARCHAR(20)	CHECK ('Scheduled','Completed','Cancelled')	Appointment status

Sample Data — Appointments

ID	PatientID	DoctorID	Date	Reason	Status
1	1	1	2026-06-15	Chest Pain	Scheduled
2	2	2	2026-06-16	Headache	Completed
3	3	3	2026-06-17	Knee Pain	Scheduled

Table: Treatments

Column	Type	Constraint	Description
TreatmentID	INT	PK IDENTITY	Auto-generated ID
AppointmentID	INT	FK -> Appointments	Linked appointment
TreatmentNM	VARCHAR(100)	-	Treatment name
Description	VARCHAR(300)	-	Details
TreatmentDate	DATE	-	Date of treatment
Cost	DECIMAL(10,2)	CHECK (Cost >= 0)	Treatment cost

Sample Data — Treatments

ID	ApptID	Name	Description	Date	Cost
1	1	ECG Test	Heart monitoring	2026-06-15	500.00
2	2	MRI Scan	Brain imaging	2026-06-16	1200.00
3	3	X-Ray	Knee bone scan	2026-06-17	300.00

Table: Prescriptions

Column	Type	Constraint	Description
PrescriptionID	INT	PK IDENTITY	Auto-generated ID
TreatmentID	INT	FK -> Treatments	Linked treatment
MedicineName	VARCHAR(100)	-	Medicine name
Dosage	VARCHAR(50)	-	Dose amount
Frequency	VARCHAR(50)	-	How often
Duration	VARCHAR(50)	-	How long
Notes	VARCHAR(300)	-	Extra instructions

Sample Data — Prescriptions

ID	TrtID	Medicine	Dosage	Frequency	Duration	Notes
1	1	Aspirin	100mg	Once daily	7 days	After meals
2	2	Paracetamol	500mg	Twice daily	5 days	If pain persists
3	3	Ibuprofen	200mg	Thrice daily	3 days	Take with food

Table: MedicalRecords

Column	Type	Constraint	Description
RecordID	INT	PK IDENTITY	Auto-generated ID
PatientID	INT	FK -> Patients	Linked patient
RecordDate	DATE	-	Date of record
Description	VARCHAR(300)	-	Record details
Diagnosis	VARCHAR(200)	-	Medical diagnosis
Notes	VARCHAR(300)	-	Extra notes

Sample Data — MedicalRecords

ID	PatientID	Date	Diagnosis	Notes
1	1	2026-06-15	Mild heart issue	Follow up needed
2	2	2026-06-16	Migraine	Reduce stress
3	3	2026-06-17	Ligament strain	Physiotherapy required

Table: Bills

Column	Type	Constraint	Description
BillID	INT	PK IDENTITY	Auto-generated ID
PatientID	INT	FK -> Patients	Linked patient
AppointmentID	INT	FK -> Appointments	Linked appointment
BillDate	DATE	DEFAULT GETDATE()	Date of bill
TotalAmount	DECIMAL(10,2)	CHECK (>= 0)	Total charged
PaymentStatus	VARCHAR(20)	CHECK ('Paid','Unpaid')	Payment status

Sample Data — Bills

ID	PatientID	ApptID	Total Amount	Payment Status
1	1	1	500.00	Unpaid
2	2	2	1200.00	Paid
3	3	3	300.00	Unpaid

View: vw_PatientHistory

Combines Patients, Appointments, Doctors, and Treatments into one easy-to-read view.

FullName	AppointmentDate	DoctorNM	TreatmentNM	Cost
Khaled Mostafa	2026-06-15	Dr. Ahmed Ali	ECG Test	500.00
Mona Adel	2026-06-16	Dr. Sara Mohamed	MRI Scan	1200.00
Youssef Tarek	2026-06-17	Dr. Omar Hassan	X-Ray	300.00

Stored Procedure: AddPatient

Inserts a new patient into the Patients table. Accepts FullName, Gender, BirthDate, and Phone as parameters.

Parameter	Type	Description
@FullName	VARCHAR(100)	Patient full name
@Gender	VARCHAR(10)	Male or Female
@BirthDate	DATE	Date of birth
@Phone	VARCHAR(20)	Phone number